

Radiometric Control Witness Form & Diagrams

Project Name: Chesterton Windmill, Windmill Hill Lane, Warwickshire

Date: 09/08/2025

Project Code: WIND

	Details	Notes/Comments	Justification/Explanation
Survey Overview			
Date & Time:	09/08/2025 08:00-09:00	Date and time of survey. Time is important	
General Survey Conditions:	Sunny, sunrise coming from 05:39 (azimuth 98° at 8:45), no visibility issues, no cloud cover, clean air	Brief description of survey conditions (inc., live and recent weather, visibility, cloud cover, air quality.	
No. of Points	22 (11 markers with two readings per marker). THIS IS REFLECTED IN THE TRI MEASUREMENTS PER POINT	Number of points being taken. Distribution of control and check points	
Equipment:			
Make / Model:	RS PRO RS-8876 InfraRed Thermometer	Make/model of radiometric equipment used.	Sport ratio of 1:20
Specifications:	Accuracy: ±1% of reading (20°C to 300°C range) Sensitivity: 0.1°C	Critical specifications only (inc., temp. range, accuracy, sensitivity, recorded metrics, spot ratio)	
Additional Equipment	Board with TESTO emission tape	Any additional equipment needed for the execution of the survey.	Board used for emissivity calculations.
Survey Team:			
Planned / Set Up by:	Neil Sutherland	Lead individual responsible for survey planning	
Observed by:	Steve Sutherland	The individual taking the point observations. This should remain consistent during the survey.	Steve Sutherland recorded values whilst readings were taken by Neil Sutherland

Computed by:	Neil Sutherland	The team member responsible for the processing of the observations	
Environmental Conditions:			
Air Temperature (AT)	21.2°C	Taken with RS Pro RS-91 Hygrometer	
Reflected Temperature (RT)	21.9°C	Taken with RS Pro RS-8876 InfraRed Thermometer	
Relative Humidity (RH)	49.7%	Taken with RS Pro RS-91 Hygrometer	
Emissivity:			
Observed Materials & Emissivities	Sandstone	Listed the materials observed and their emissivities. This includes any values determined outside of this survey.	
Observation Method	ASTM E1933-14	Brief description of how emissivity was observed and calculated (inc., no. of observations and redundancy)	ASTM Emissivity standard used
Notes:			
General Notes:	Temperature measurements were taken perpendicular on the thermal-specific survey markers (adhered with blue tac) Two measurements taken on rubber elements making up cross hair: the top right square (Rubber_TL) and bottom left square (Rubber_BR) Additional measurements taken on aluminium element (Aluminium_TR)		

POINT 1

	Details	Notes/Comments	Justification/Explanation
Point Name:	T1		
Temperature (°C):	Marker (Rubber_TL): 25.3 Marker (Rubber_BR): 26.4 Marker (Aluminium_TR): 22.2		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	

Diagrams

**Control Point
Photo(s):**



POINT 2

	Details	Notes/Comments	Justification/Explanation
Point Name:	T2		
Temperature (°C):	Marker (Rubber_TL): 20.4 Marker (Rubber_BR): 20.2 Marker (Aluminium_TR): 19.1		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	

Diagrams

**Control Point
Photo(s):**



POINT 3

	Details	Notes/Comments	Justification/Explanation
Point Name:	T3		
Temperature (°C):	Marker (Rubber_TL): 18.1 Marker (Rubber_BR): 17.8 Marker (Aluminium_TR): 20.0		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	Marker under the blades of the windmill, to the LEFT of the lightning wire (as you look at it)

Diagrams

**Control Point
Photo(s):**



POINT 4

	Details	Notes/Comments	Justification/Explanation
Point Name:	T4		
Temperature (°C):	Marker (Rubber_TL): 17.0 Marker (Rubber_BR): 16.6 Marker (Aluminium_TR): 18.0		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	Marker under the blades of the windmill, to the RIGHT of the lightning wire (as you look at it)

Diagrams Control Point Photo(s):	
	

POINT 5

	Details	Notes/Comments	Justification/Explanation
Point Name:	T5		
Temperature (°C):	Marker (Rubber_TL): 18.5 Marker (Rubber_BR): 20.7 Marker (Aluminium_TR): 15.8		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	Next marker along from T4, next column along (whereas others are separated by a full column)

Diagrams

**Control Point
Photo(s):**



POINT 6

	Details	Notes/Comments	Justification/Explanation
Point Name:	T6		
Temperature (°C):	Marker (Rubber_TL): 21.9 Marker (Rubber_BR): 23.8 Marker (Aluminium_TR): 22.5		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	Next column along from T3 and T4 (whereas others are separated by a full column)

Diagrams

**Control Point
Photo(s):**



POINT 7

	Details	Notes/Comments	Justification/Explanation
Point Name:	T7		
Temperature (°C):	Marker (Rubber_TL): 34.6 Marker (Rubber_BR): 31.9 Marker (Aluminium_TR): 32.2		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	Column next to millstone, either side of the lightning rod NON-blade side. In direct sunlight during survey!

Diagrams

**Control Point
Photo(s):**



POINT 8

	Details	Notes/Comments	Justification/Explanation
Point Name:	T8		
Temperature (°C):	Marker (Rubber_TL): 31.0 Marker (Rubber_BR): 31.7 Marker (Aluminium_TR): 26.8		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	Column next to millstone, either side of the lightning rod NON-blade side. In direct sunlight during survey!

Diagrams

**Control Point
Photo(s):**



POINT 9

	Details	Notes/Comments	Justification/Explanation
Point Name:	I2		
Temperature (°C):	Marker (Rubber_TL): 20.1 Marker (Rubber_BR): 22.4 Marker (Aluminium_TR): 23.2		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	'I' stands for INTERIOR (I2, I3 or I12)

Diagrams	
Control Point Photo(s):	 The image consists of two side-by-side photographs of a stone pillar base. The left photograph is a color image showing a stone pillar base on a cobblestone floor. A red circle highlights a small metal pin or marker on the base of the pillar. The right photograph is a grayscale version of the same scene, also with a red circle highlighting the same pin. The background shows a landscape with fields and a blue sky with clouds.

POINT 10

	Details	Notes/Comments	Justification/Explanation
Point Name:	I3		
Temperature (°C):	Marker (Rubber_TL): 20.0 Marker (Rubber_BR): 20.3 Marker (Aluminium_TR): 24.7		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	'I' stands for INTERIOR (I2, I3 or I12)

Diagrams	
Control Point Photo(s):	 The image consists of two side-by-side photographs of a stone pillar base. The left photograph is a color image showing a stone pillar base on a cobblestone path, with a red circle highlighting a small mark. The right photograph is a grayscale image of the same scene, also with a red circle highlighting the same mark. The background shows a landscape with fields and a blue sky with clouds.

POINT 10

	Details	Notes/Comments	Justification/Explanation
Point Name:	I12		
Temperature (°C):	Marker (Rubber_TL): 24.5 Marker (Rubber_BR): 22.3 Marker (Aluminium_TR): 22.7		
Emissivity (ϵ):	Marker (Rubber_TL): 0.91 Marker (Rubber_BR): 0.91 Marker (Aluminium_TR): 0.37		
Material:	Rubber Rubber) Aluminium		

Notes:	
Notes:	'I' stands for INTERIOR (I2, I3 or I12)

Diagrams	
Control Point Photo(s):	